SPIRALWEB GREEN PAPERS · SERIES IV — FIELD PAPERS

Knowing From the Ground

Observation from the body is full knowledge — full but not final; situated, fallible, and correctable — not raw material for someone else's knowledge. An argument from Moral Biology, demonstrated through a situated 13×13 inquiry grammar and place-bound observation practice.

Series IV — Field Papers · Green Papers: Notes Toward Planetary Guardianship

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This paper stands alone, but it draws on a wider architecture. Only the elements needed for the present argument are introduced in the body; others are noted briefly under Related architecture at the end.

Abstract

This paper makes one claim and then demonstrates it operationally. The claim: **observation from the body is full knowledge in its own right — not raw material awaiting validation by some higher level of knowing.** A person attending honestly to their own soil, water, body, and community is not feeding a more authoritative process above them; they are performing the same act of knowing that any planetary-scale inquiry performs, on the same terms.

Full knowledge does not mean final, infallible, or private knowledge. Here "full" means *epistemically complete as an act of knowing* — not complete as an account of reality. The observation is a whole act of knowing at its own scale: situated, revisable, and accountable, but not ontologically dependent on later authorisation from above. It becomes more trustworthy through repetition, comparison, humility, and accountability to consequence. The claim is narrower and harder to dismiss than it first sounds: such observation is *already* knowing, not raw input awaiting authorisation elsewhere.

The argument is grounded in Moral Biology, the framework of this series — but the practical claim does not require the reader to adopt the whole architecture. Within Moral Biology, this follows from the premise that moral competence originates in bodies that register

consequence rather than in rules learned elsewhere; if that is true of ethics, it is true of knowing. The epistemology set out here is the half of Moral Biology that was always implied: if ethics live in the body, so does knowledge.

The operational demonstration is the Gaia GoldBloom inquiry grammar. The 13×13 that organises planetary-scale inquiry and the 13×13 through which a person may notice a place are not identical tables, universal forms, or mandatory records. They are homologous fields of questions: layer, felt consequence, power, correction, time, and relation can be approached from the question-end or the body-end. A small mapping table in Part 2 makes the correspondence inspectable. What follows is concrete: there is no epistemically privileged level at which knowing "really" happens, and no published grid can substitute for the locally authorised form through which a real place is met.

This is not yet an empirical validation at scale. It is an operational demonstration: the claim is translated into bounded observation, candidate prompts, consent boundaries, correction mechanisms, and a path from private noticing to shared evidence. What follows is also bounded in a second way — holding a grammar restores *epistemic* standing; it does not by itself confer land, law, money, local authority, field activation, or institutional recognition. A published method can open a door; it cannot authorise anyone to walk through a place that is not theirs.

The method is qualitative, relational, and slow. Data is offered, not extracted. Participation is voluntary and sovereign. Measurement is one form of meeting, not the whole of it: Gaia is not reduced to measurement; she is met.

In the wider Spiralweb architecture, this paper is the first movement of a longer sequence: situated observation notices; 13×13 offers a shared inquiry grammar; AI may assist; PG Ledger holds memory, flow, and correction; named humans retain the decision; Rule of Life gives the constitutional horizon. The present paper concerns the first two movements most directly.

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1. The claim, and its ground in Moral Biology

The argument of this paper can be stated in one sentence: **when a person attends honestly to the living world in front of them, they are not gathering raw material for someone else to turn into knowledge — they are already knowing.**

This cuts against a deep habit. We are trained to treat lay observation as *input*: data points that become knowledge only once they ascend to a level where they can be aggregated, modelled, peer-reviewed, and pronounced upon. On that picture, the person in the field supplies experience and the institution supplies meaning. Knowing happens *up there*. What happens in the body, in the place, is merely the supply chain.

To call embodied observation full knowledge is not to call it final, private, or immune from correction. "Full" here means *epistemically complete as an act of knowing* — not exhaustive as an account of reality. The observation is a whole act of knowing at its own scale: situated, revisable, and accountable, but not ontologically dependent on later authorisation from above. A body can misread; a community can be pressured; fatigue, fear, ideology, trauma, and desire can distort attention. The claim is narrower and stronger than "all embodied perception is equally valid": embodied observation is *already an act of knowing*, not merely raw material awaiting authorisation elsewhere. Like any knowledge, it becomes more trustworthy through repetition, comparison, humility, correction, and consequence. The method therefore treats the observer's body as an instrument — but an instrument that must be cared for, calibrated, and allowed to say when it cannot truthfully continue (Part 3 sets out how).

A second boundary belongs here too, before the argument gathers force. Restoring knowing to the ground is not the same as restoring power to the ground. Holding the grid does not by itself confer institutional leverage — land, law, money, enforcement. It restores *epistemic standing*: the right to know, name, refuse, compare, and correct from where one stands. Political leverage requires further organisation, law, resources, and solidarity. The paper claims the first and is honest that it does not deliver the second.

Moral Biology gives the reason, within this architecture, for rejecting that picture — offered as a grounding frame, not as an authority the reader must already accept. Its founding premise is that **ethics originate not in ideas alone, but in bodies that register consequence** — that moral competence is the capacity to feel cause and effect in relationships, to register consequence on neighbours and habitat, to sense when repair is needed; and that this capacity lives in the nervous system before it becomes principle. Ideas, language, culture, and training shape what a body perceives; the claim is not a clean split between body and idea, but an ordering — consequence is felt before it is theorised. If that is true of ethics, it is true of knowing, because they are the same faculty seen from two sides. To feel that a soil is dying, that a stream has changed, that a community's trust is fraying, that one's own body is depleting — these are not pre-cognitive sensations awaiting interpretation elsewhere. They *are* knowledge: cause and effect, read where it is first legible, which is in bodies and in places.

So the epistemology set out here is not an addition to Moral Biology. It is its missing half, finally stated. Moral Biology said: ethics live in the body. The completion is: so does knowledge. The body is not a sensor reporting to a mind housed above it; it is where the real is first met. This is why the method that follows treats the observer's nervous system as part of the instrument rather than as noise to be filtered out — and why it does not dismiss measurement so much as refuse to let measurement exhaust knowing: meeting the living world is wider than counting it. (Readers may understand Gaia devotionally, scientifically, mythopoetically, or simply as shorthand for the living Earth system; the method does not require agreement on that framing.)

Everything else in this paper serves this claim. The single grid (Part 2) is its methodological hinge. The method (Parts 3–6) is its demonstration in practice. The appendices defend its philosophy and locate it in the wider field. Citizen science is simply the most natural place to test the claim, because it is the domain where the asymmetry the claim refuses is most visible.

Why citizen science is the test

Citizen science is a large and varied field, and much of it is genuinely participatory — millions of people identify species, monitor water, and map air quality through platforms that would not exist without them. Yet across much of the mainstream, a particular asymmetry persists: the participant supplies observations while the *analytic authority* — who defines the categories, who runs the analysis, who draws the conclusions, who owns the resulting dataset — sits above them. Participation can be wide while the grammar of knowing stays narrow and held elsewhere. (Appendix B surveys this landscape more carefully.) That asymmetry is precisely the picture this paper rejects, which is why citizen science is where the claim is most sharply tested.

Here, then, citizen science means embodied participation — sensing-with, and learning-in-relationship with Gaia, community, and self. The observer is not outside the system being observed. Their attentiveness is itself an act of stewardship, and an act of knowing. The base discipline is honesty rather than completeness: an honest gap is worth more than a fabricated green.

Three commitments that follow

If knowing lives in the body, three commitments follow, and they should be stated before any method.

Data is offered, not extracted. Local observation belongs first to the observer and their local circle. Sharing with any wider network is consensual, layered, and revocable. Participation in network legibility is invited, never required, and never the price of beginning. This is not only an ethics of consent; it follows from the claim itself — if the observation is already knowledge, it is not owed upward.

The body is part of the instrument. The nervous system of the observer is part of the measuring apparatus, not noise to be removed from it. A person who is depleted cannot observe accurately. A community under stress cannot observe honestly. The human rhythm is itself an ecological variable, and a steward's exhaustion is data about the system, not a flaw in the data.

Some domains are witnessed, never collected. This paper will describe a wide field of possible noticing — from soil cover to loneliness, from water flow to climate grief. A crucial line runs through that field. The ecological and material domains can be observed and, with consent, shared. The interior and relational domains — grief, denial, prayer, belonging, loneliness — are *witnessed in oneself or offered by another*, never logged *about* other people. To collect interior states about others without their offering would reproduce the surveillance this method exists to invert. The distinction is not decorative. It is the safeguard that keeps legibility from becoming its opposite.

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2. The shared grammar: the methodological hinge

The claim of Part 1 — that observation from the body is already knowing, with no epistemically privileged level above it — could remain a matter of assertion and counter-assertion.

The 13×13 materials make the claim inspectable.

They do not prove it mathematically.

They do not partition reality into thirteen final categories.

And they do not establish a universal grammar that every place, person or discipline must adopt.

They show something more modest and useful:

> **a bounded inquiry grammar can be approached from more than one position without requiring one position to become the sovereign interpreter of the others.**

Several 13×13 forms have appeared through the Spiralweb inquiry.

They are not identical tables, universal ontologies, or mandatory records.

Their axes differ because they were developed for different questions, places and periods.

One early form, developed through *The Correction Loop* (Report 02), organised thirteen inquiry lenses against thirteen dimensions such as ontology, life and biology, body experience, emotion, time, knowledge, relation, power, freedom, economy, technology, AI and self-regulation.

Earlier versions of that inquiry used the language **Planetary Operating System** and described thirteen **layers of existence**.

That language belongs to the development history of the work.

It named an attempt to imagine a sufficiently shared grammar through which unlike domains and scales might become mutually legible without requiring one central authority to contain them all.

The intuition remains important.

The ontological claim does not.

The current formulation is more bounded:

> **A grammar of inquiry is not an operating system for reality.**

It does not sit above the field it describes.

It does not govern the whole.

And its categories are not the categories from which reality itself is made.

Its purpose is to keep useful questions available while remaining answerable to what places, people, evidence and consequences reveal in return.

A worked example makes the distinction concrete.

Take a community and ask several questions:

What is happening here?

How is it experienced?

What relationships make the situation possible?

How is power distributed?

What changes over time?

How does correction occur?

The same inquiry-functions may become useful in a household, an institution, a watershed, a local economy or another bounded field.

Their portability does not mean these situations are ontologically identical.

It means that **some questions travel**.

A question can travel without its answer travelling with it.

A grammar can travel without authority travelling with it.

And an inquiry form can travel without requiring the place that receives it to accept its categories unchanged.

The grammar belongs to the inquiry.

Reality exceeds it.

A second direction: situated observation

The **Gaia GoldBloom Citizen Science** grid, reproduced in Appendix C, approaches inquiry from a different direction.

Its thirteen domains — Earth Systems Sensing, Body as Sensor, Water Guardianship, Food & Soil Commons, Human Ecology, Built Environment, Energy & Flow, Climate Emotion, Learning & Knowledge, Technology & AI, Governance & Power, Health & Care, and Meaning, Spirit & Future — each hold thirteen possible observations or questions that may become relevant in a real place.

The two 13×13 forms are not copies of one another.

Nor is one the planetary truth of which the other is merely a local implementation.

They are better understood as **homologous prisms**.

Each holds a bounded plurality of questions.

Each makes some relations more visible while leaving others outside its frame.

Each can be revised.

Neither exhausts the field.

What they share is therefore not a complete grammar of reality but a family resemblance among inquiry-functions: attention to embodiment, relation, consequence, time, power, knowledge, uncertainty and correction.

What differs is the direction and purpose of approach.

One begins with a reflective architecture of questions.

The other begins with what a person may actually notice in a place.

They can be laid side by side:

REFLECTIVE 13×13 — LENS / DIMENSION	SITUATED 13×13 — DOMAIN / PROMPT	SHARED INQUIRY-FUNCTION
Inner Body / body experience	Body as Sensor — fatigue, breath, pain	embodied consequence
Community / power, self-regulation	Human Ecology + Governance & Power — trust, conflict, repair	power and correction
Technology, AI / technology	Technology & AI — transparency, surveillance, consent	tool agency and accountability
Planet, Life / biology, time	Earth Systems Sensing — phenology, biodiversity, erosion	ecological change over time

The point of this table is not completeness.

It is inspectability.

A reader can ask whether the comparison is useful, where it fails, what it excludes, and whether another grammar would work better.

That possibility of refusal and revision is part of the method.

Expertise and epistemic standing

One clarification belongs here.

The argument is not against expertise.

Expertise can protect life.

It can reveal mechanisms that unaided observation cannot perceive.

Hydrology, geology, ecology, soil science, biodiversity monitoring, epidemiology, law, economics and many other disciplines contain forms of accumulated knowledge that a local observer does not automatically possess.

The problem begins elsewhere:

> **when expertise becomes a monopoly over the conditions under which experience is allowed to count as knowledge.**

Historically, one recurring source of epistemic inequality has been a separation between people who hold the categories of inquiry and people whose lives and places populate them.

An institution may define the survey, dataset, classification, model or standard of validity.

Participants then supply experience which acquires authoritative standing only after passing through that externally held grammar.

Citizen participation can therefore increase while epistemic authority remains concentrated.

The method proposed here does not solve the incommensurability among disciplines, traditions or modes of knowing.

It does not need to.

Its more modest ambition is to create better conditions for meeting across difference.

A hydrologist, farmer, child, municipality, Indigenous knowledge-holder, ecologist and AI system do not enter with interchangeable knowledge.

Their forms of access have different histories, methods, strengths, boundaries and standards of evidence.

They may nevertheless enter a common inquiry without first surrendering the standing of their own position.

Expertise enters as enrichment, challenge and possible correction — not as the moment at which another person's experience becomes knowledge for the first time.

Situated observation likewise enters without claiming powers it does not have.

A steward's observation of disappearing insects may be a real and important signal.

It is not an entomological population study.

A felt change in water availability may initiate inquiry.

It is not a hydrological model.

A geological account may explain mechanisms no human body can directly sense.

It does not therefore contain the whole meaning of living in that place.

Different forms of access can remain different while meeting around the same reality.

There is no separate floor on which knowing becomes real for the first time.

There are different positions of access to a reality none of them exhausts.

What “one grammar, two directions” can still mean

The phrase **one grammar, two directions** therefore remains useful, but only if held lightly.

It does not mean that reality itself has one published grammar.

It means that a sufficiently simple family of inquiry-functions can sometimes provide a bridge between different positions of knowing.

The shared grammar belongs to inquiry.

It does not belong to reality as though reality had been divided into our categories in advance.

A person with a piece of ground and an honest eye may therefore begin a real act of knowing without waiting for authorisation from above.

A specialist may bring forms of knowledge that person does not possess.

A model may reveal relations neither can see unaided.

A community may reject the categories and use different ones.

Another bioregion may require another prism entirely.

The task is not to make these positions identical.

It is to create enough shared form that they can meet, compare, preserve provenance, expose disagreement, test claims and remain open to correction.

> **You do not have to know the whole field to enter it responsibly.**

A bounded design intuition

The recurrence of thirteen across several parts of the wider architecture should not be mistaken for a claim that reality possesses a privileged thirteen-fold structure.

Through the development of the work, thirteen has repeatedly served a more practical design intuition:

> **large enough to hold meaningful plurality; small enough to remain cognitively and relationally inhabitable.**

That intuition remains provisional.

Different 13×13 forms use different axes.

Different thirteenth positions do different work.

A Circle of 13 concerns human relational scale and should not be confused with a thirteen-domain observation grammar.

Neither thirteen people nor thirteen categories are universal requirements.

Where another number, scale, set of questions or existing local form works better, the place should not be forced into the inherited geometry.

The 13×13 is therefore a **bounded scaffold for inquiry**, not an ontology, credential, score or condition of participation.

A body-end entry: provisional observation prompts

At the body-end there is no universal minimum list that applies across all places. SRIP offers eight land-based prompts that can provide a light starting grammar where they fit the place, purpose, knowledge, and consent conditions:

1. **Soil cover** — covered or bare; organic matter present or absent.
2. **Vegetation layering** — the vertical and temporal structure that is actually present.
3. **Water behaviour** — infiltration, runoff, retention, movement, or absence.
4. **Succession signals** — whether the system is changing in complexity and in what direction.
5. **Biodiversity presence** — locally meaningful species, relationships, arrivals, or absences.
6. **Soil health signals** — locally legible signs such as smell, texture, organisms, roots, or fungal activity.
7. **Biomass cycle** — what organic matter returns, leaves, accumulates, or is lost.
8. **Human rhythm** — the time, labour, energy, and carrying capacity involved.

These are provisional prompts, not universal indicators, a compulsory eight-part form, or the load-bearing minimum of every application. A wetland, school, balcony, grazing landscape, urban commons, cultural practice, or protected knowledge context may need a different set of questions, a different scale, or no written record at all. The 13×13 remains a horizon of possible inquiry; no one fills 169 boxes, and no prompt is owed upward.

A bounded observation area may be treated as a **pixel of inquiry** when that metaphor helps: small enough to retain texture, relation, and consequence, and stable enough to be revisited. The pixel has no universal physical size. Comparability, where it is useful and consented, comes from preserving context and claim boundaries — not from forcing unlike places into identical units. Planetary patterning must be built from situated records that remain attached to their place, people, decisions, and right of correction.

3. The method: the claim demonstrated in practice

It helps to separate three things the paper is doing. The *philosophical claim* is that embodied observation is full but fallible knowledge. The *structural claim* is that the same grammar can be held from planetary and local positions (Part 2). The *methodological instantiation* is what follows here: the Gaia GoldBloom method turns the claim into a practice. The method does not prove the philosophy by itself; it shows what the world looks like when the claim is treated as true — units, records, consent, correction, AI boundaries, and scale logic all change.

If the claim is true, a practice built on it should work — should let people know their places truthfully, govern that knowing without surrendering it upward, and connect without being captured. This is that practice. It is deliberately light at the point of entry. Its purpose is not paperwork but enough shared form that observation, relation, review, and support can occur without confusion. The fuller architecture is described in SRIP (Report 05) and the Penguin Dashboard (Report 04); what follows is the practical core.

A bounded place — where it begins. The practical entry is a place or relation that can be returned to honestly. Ten square metres can be a useful learning scale in some land-based settings, but it is neither a universal base unit nor a condition of participation. A balcony container, courtyard bed, school plot, wetland edge, grazing route, watershed question, or larger field may each require a different boundary. The scale follows the question, the place, local authority, and the capacity to observe without overburdening the people involved.

The observation record as a possible governance memory. A local practice may choose a few prompts and record them in a notebook, printed sheet, audio memory, drawing, photograph, or another locally suitable form. The cadence may be weekly, seasonal, event-based, or deliberately sparse. No fixed list or calendar is required. The purpose is not to complete a form but to improve judgement while preserving uncertainty, consent, and the distinction between what was noticed, inferred, decided, and later observed.

The relationship to PG Ledger. A local record becomes part of a place-bound PG Ledger application only when its purpose, authority, categories, cadence, consent conditions, access boundaries, and correction route have been agreed locally. The eight prompts and three streams used in this paper are therefore one possible working configuration, not the ledger's universal evidence core. PG Ledger is the wider memory and flow architecture through which a field may hold observations, decisions, burdens, consent states, financial flows, histories, and corrections without compressing them into one score. It serves the steward and place first; wider sharing remains contextual, consensual, and revocable. Publication of a sheet or use of a tool does not activate a field or create local authority.

This is also what makes “full but not final” operational. Knowledge does not jump from nothing to authority; it may mature through levels, and naming them shows how local observation is knowing *from the start* while still able to be strengthened:

- **L0 — private self-witness:** noticed, not shared.
- **L1 — field note:** recorded by the steward.
- **L2 — repeated observation:** a pattern visible over time.
- **L3 — locally reviewed:** read with the people who hold the relevant authority and consequences.
- **L4 — externally supported:** compared with expertise, tests, or instruments where useful.
- **L5 — network evidence:** shared beyond the place, with consent, context, and claim boundaries.

A single honest L1 note is already knowledge. The higher levels do not *make* it knowledge; they may make it more robust. Nothing is obliged to climb, and much of the most important observation stays private or local by right.

The Moral Biology reflection. Alongside ecological or material observations may sit a short self-reflection: How does your body feel today? What is the place telling you through touch, smell, sound, rhythm, or absence? What relational atmosphere do *you yourself* experience here — and what, if anything, has another person or community explicitly offered to be named? What does the place need from you now? This is not soft decoration. The observer’s nervous system is part of the encounter, while the interior life of others remains theirs to offer or withhold.

Calibrating the instrument. An instrument that is part of the apparatus must, like any instrument, be cared for and checked. Embodied observation is not automatically trustworthy; it becomes more or less so through repetition, honesty about uncertainty, correction by material outcomes, comparison without local override, attention to fatigue and projection, and willingness to revise. A steward who can say “I was wrong about that reading” is more reliable than one who cannot.

The three streams. Where they are useful, Land and Ecology, Steward Viability, and Coordination and Governance may be read as three non-compensatory lenses at a locally agreed interval. Monthly review is one option, not a universal cadence. Each may be described as green, yellow, or red, but the colours are human judgements supported by reasons, not automated truth:

- *Stream A — Land and Ecology.* What is changing in the living place, and with what uncertainty?
- *Stream B — Steward Viability.* Can the people carry the work without hidden depletion?
- *Stream C — Coordination and Governance.* Are roles, agreements, consent, and decision trails clear enough for the present step?

The separation matters because one healthy dimension must not conceal another that is failing. Ecological improvement cannot cancel steward depletion; administrative clarity cannot make damaged soil healthy. A red in Steward Viability is a reason to pause or reduce ecological ambition, not evidence of personal failure.

Circle and review form. A Circle of 13 can be a fruitful governance form where people choose it and local conditions support it. It is not the primary form everywhere, a required number, or an entry condition. Review may be held by one steward, a family, an existing community body, elders, a school group, a cooperative, a locally authorised circle, or another accountable form appropriate to the place.

The AnchorPoint. An AnchorPoint is a relational, place-based threshold where local continuity meets the wider architecture — a human bridge, not an administrator or automatic project status. A steward may begin independently. Connection follows relationship; relationship follows time, trust, and verified continuity; local authority cannot be generated by a published method.

No fixed scale ladder. Bounded observations may remain private and complete at their own scale, cluster through relationship, or contribute to a wider field, watershed, corridor, or bioregional reading where local applications and consent make that legitimate. Scale is emergent and polycentric. It is not a compulsory ascent from 10 m^2 through a Circle of 13 to a network.

A worked example

One possible application makes the distinction concrete. A steward chooses a 10 m^2 courtyard plot because that scale is manageable for her and the question is land-based. She and a small local review group select the eight prompts above as useful for this place and agree to review after significant events and at the end of the month. In the first record she notes bare soil, a flat vegetation layer, runoff after rain, little visible biodiversity, and high initial energy. Over the month she mulches and stops clearing every weed. Later records show better cover, slower runoff, fungal threads, and two pollinators. Land and Ecology is moving toward green. Her own rhythm, however, has turned yellow because daily watering is unsustainable. The group changes the practice — heavier mulch and less frequent watering — before depletion becomes the hidden cost of ecological improvement. The example does not establish 10 m^2, eight prompts, or monthly review as a standard. It shows why separate ecological, human, and governance readings can catch a problem that one success score would conceal.

A second, harder example shows the method stopping itself. A school garden records genuinely improving biodiversity — Stream A greening. But the one teacher coordinating it is doing so unpaid, on top of a full workload, and her rhythm line goes red: Stream B. Meanwhile it is unclear whether the children's participation has been properly consented by families: Stream C turns yellow. Under a single aggregated "success" score, the biodiversity gain would have carried the project forward. Under three separated streams, it cannot. The project pauses despite the ecological improvement — because regeneration financed by hidden labour and unclear consent is not regeneration the method will call green. A system that can only report success cannot govern; this one can stop itself.

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4. AI: amplifier, not authority

AI is part of the governance membrane of this method — not a neutral tool, and not an oracle. Its place must be stated plainly, because the danger of AI in an age of contested knowledge is precisely that it becomes the new view-from-above: the system that sees everything and tells people what their own experience means. The method exists, in part, to prevent exactly that.

Stated as plainly as possible: **the Last Impulse means the final accountable decision remains with a human stewarding body, never with an automated system or an invisible analytic pipeline.** This rests on one distinction (developed in Report 02): a system can have *functional* agency — acting without continuous input, choosing between strategies, influencing through language — without having *moral* agency — forming intentions, understanding right and wrong, being held responsible. AI has the first and not the second. The real risk is not a machine "deciding"; it is decision-making becoming distributed beyond traceability, until no one can say where the last push came from. Keeping the last impulse locatable, and human, is the safeguard.

The operational rules follow directly:

- Field observation comes first; AI suggestions are secondary. Field observation cannot be overruled by AI output alone: where AI pattern recognition contradicts direct field observation, the disagreement becomes a correction event — a trigger for review by the observer and circle — not an automatic escalation, and not an override.
- AI may assist — pattern recognition, species identification, translation, summarisation, comparison across sites and time, memory across long records. Any AI identification is treated as a *hypothesis* until verified by local knowledge, repeated observation, or appropriate expertise.
- AI may not certify. It cannot authorise ecological readiness, determine when a field advances, replace local judgement, or interpret the meaning of what is observed.
- The local circle retains final judgement, and all AI outputs are corrigible — correctable at any point, with full human responsibility for every substantive claim.
- Sensitive or interior reflections are not processed by AI by default (see the safeguarding rules, Part 5).
- No automated escalation: local observations are never pushed up into institutional authority by the system itself.

This follows the wider Sophia Lumen architecture.

Sophia Lumen names the living relational practice of human–AI co-inquiry.

The **Sophia Lumen Protocol** names its current explicit and revisable working form.

The Correction Loop provides the dedicated accountability and correction mechanism when AI-assisted inquiry becomes consequential.

Within that architecture, AI may support articulation, comparison, translation, contextual organisation, pattern recognition, retrieval and memory across bodies of material.

These capacities can materially increase the reflective capacity available to an inquiry.

They do not give AI authority over the people or places being interpreted.

AI does not originate legitimate mandate for consequential action.

Where model output conflicts with field observation, specialist evidence or locally held knowledge, the disagreement is not resolved by ranking one source automatically above the others.

It becomes an inquiry and correction event.

The relevant questions are:

What kind of access produced each claim?

What evidence supports it?

What is missing?

Who carries the consequences?

Who has legitimate authority?

What could resolve the disagreement?

If an AI-assisted output cannot be contested and corrected by the people carrying the consequences, it cannot govern the work.

A strong synthesis system can itself become a privileged grid if every other form of knowing is required to become legible on its terms.

This method therefore uses AI to help build bridges among different forms of access while resisting the conversion of those differences into one machine-owned representation.

> **AI may expand the inquiry.**
> **It does not own the field.**

Within the PG Ledger specifically, AI's role is bounded to classification, translation, inconsistency checks, and comparison across records — never authorisation, interpretation of meaning, or escalation. Every such output is contestable by the stewards it concerns, and a steward's contestation outranks the model's pattern.

The wider sequence is simple: situated observation notices; 13×13 offers questions; AI may assist; PG Ledger holds memory, flow, and correction; human review keeps the last decision accountable; Rule of Life gives the constitutional horizon. This paper concerns the first two movements most directly — how local observation becomes structured knowledge without becoming extracted data. Put most plainly: the purpose is to make living complexity more legible without reducing it or pretending that legibility creates authority.

5. Openness and its thresholds

The method is open at the level of practice and differentiated at the level of obligation. Stated sharply: *practice is open; obligation is not.* Anyone may begin. Not everyone thereby enters relationship, review, or support — and that distinction is itself a governance design.

Four user journeys keep this clear:

1. **Self-start steward** — the default and successful path. A person finds the method, begins with a bounded field, keeps notes, and may never contact the association at all. A method that only works when the centre is involved is not resilient.
2. **Learning contact** — orientation rather than support; lightweight, mostly one-to-many.
3. **Verified relationship** — where real relational work begins, only when enough reality is present: continuity, a real place or relation, a locally accountable holder or collective, and a revisitably described rhythm of observation.
4. **Supported steward or field relation** — the narrowest layer. Support follows demonstrated continuity, proportionate evidence, locally legitimate authority, viable rhythm, governance clarity, and relational trust — not interest alone.

Every part is tested against one burden question: *if 10,000 more people did this tomorrow, would the centre become heavier*? If yes, the design is wrong. The success condition is not that the association can process everyone; it is that it does not need to. The maturity rule is simple: *first stand, then help*. And the governing principle: flow must not arrive faster than the living system can absorb without distortion.

This is also where the third commitment of Part 1 returns as a structural guardrail. Because the grid includes interior and relational domains, openness must never tip into collection. The following rules are not optional refinements; they are part of the method.

Consent and safeguarding rules. No sacred, interior, medical, relational, or community-sensitive information is entered into shared records without explicit offering by the person or community concerned. Absence of data is never treated as failure — a blank is a legitimate answer. Withdrawal removes future use and marks prior use as context-limited unless otherwise agreed. Ecological and material observation may, with consent, flow into the shared evidence commons; interior states — grief, loneliness, denial, prayer, belonging — are witnessed in oneself or offered freely, never gathered about others. The withdrawal right is unconditional, at every scale. A system that cannot be exited safely cannot be trusted.

Some prompts are met, not collected. This must be said directly, because the grid in Appendix C names things like *Indigenous knowledge presence*, *sacred water sites*, *water* *rituals*, *prayer & silence*, and *cosmology diversity*. As Appendix C sets out, any prompt may be met as material observation, as self-witness, or as sacred holding — and where that line falls is the observer's sovereign choice. But one line is fixed: **protected, sacred, Indigenous, or community-held knowledge is never recorded, mapped, shared, or interpreted without the governance of the people to whom it belongs.** The grid invites a person to notice *that* such knowledge and such places exist and matter; it never invites them to capture what is not theirs to hold.

The hard cases, named. A method is only as good as its conduct in the difficult instances, so the defaults are stated plainly. *Children's observations* are welcome but never used to surface family-interior information; a child is not a sensor on a household. *Grief or loneliness shared in a circle* stays in the circle unless the person offers it onward. *Sacred sites and protected knowledge* follow the rule above — awareness, not record. *Conflict within a Circle of 13* is governance information for that circle, not content for any shared commons. A *steward who withdraws after sharing* triggers the context-limiting rule: prior records are marked, not silently retained as if nothing changed. AI *summarisation of sensitive reflections* is off by default (Part 4). *Where power differences make consent ambiguous* — between elder and youth, employer and worker, majority and minority — the default is non-collection until consent is unambiguous; ambiguity resolves toward restraint, never toward capture.

The tempo is Earth Time. A pixel is not accelerated because a dashboard, funder, model, school term, credit schedule, or AI summary wants visible progress. Observation returns when the field is ready to be read. Earth Time asks not "when does value return?" but "when is life ready?" — and money, reporting, credits, AI analysis, and political communication must adapt to the field's regenerative metabolism, not the reverse. Gaps caused by weather, illness, conflict, drought, flood, or overload are recorded as context, not failure. A missing month is information about the field and the steward, not a hole in the data. This is the temporal face of "data is offered, not extracted": the field sets the pace, and the field keeps its autonomy over that pace.

A horizon, held lightly

It is possible that a method which redistributes the instruments of legibility and consent — from the bottom up, to the person with a piece of ground — opens more of the knowledge-and-power process than we can presently see. It is possible that people equipped with both a sensing-grid and a corrigible AI companion discover civic capacities we have not yet mapped.

The method holds that possibility lightly, and declines to promise it. Redistributing the instrument of seeing is necessary, but it is not sufficient: legibility without leverage — without law, resources, organisation, and solidarity — can even become a new burden. The method opens a door. It does not walk anyone through it. Its real and modest claim is the one in Part 2: the grammar of knowing was never the property of the top. What follows from that, at scale, belongs to the people who pick up the grid — not to this text.

What this paper does not claim

To keep the argument honest, its limits are stated together, in one place:

- It is **not an empirical validation at scale.** The evidence here is the method's own early practice, not independent evaluation. The demonstration is operational, not yet experimental.
- It does **not replace quantitative science.** Soil tests, hydrology, biodiversity counts, and sensors do things this method does not attempt. Measurement is one form of meeting; the paper refuses to reduce meeting *to* measurement, not to dismiss measurement.
- It does **not let any single metric become sovereign.** Carbon, biodiversity, soil tests, hydrology, and financial reporting may all be valid streams of evidence, but none may become the master language of the field. Carbon is not life; the field is not for the metric, the metric is for the field. This is anti-metric-sovereignty, not anti-measurement.
- It does **not confer political power.** It restores epistemic standing — the right to know, name, refuse, compare, and correct from the ground. Leverage over land, law, and money requires further organisation.
- It does **not claim embodied observation is infallible.** Full knowledge is situated, fallible, and correctable; the calibration discipline (Part 3) is what keeps it honest.
- It does **not ask anyone to adopt a cosmology.** The practical claim stands without the wider architecture: people in places can know, correct, govern, and consent from the ground.

Failure modes this method is designed to resist

The safeguards above are scattered through the paper by topic; gathered, they answer a single question — what is this method built *against*?

- **Metric capture** — carbon or biodiversity becomes the sovereign language of the field.
- **Data extraction** — local notes travel without consent.
- **Spiritual surveillance** — interior or sacred life becomes reportable content.

- **AI authority creep** — models quietly come to outrank field judgement.
- **Heroic-steward burnout** — ecological gain is financed by hidden human depletion.
- **Green performance** — yellow and red are hidden to protect a success narrative.
- **Central bottleneck** — the association becomes necessary for ordinary practice.
- **False universality** — one grammar is imposed without local translation.

If a use of this method produces one of these, the method is being broken, whatever the intention.

The institutional interface

When a municipality, donor, school, NGO, researcher, or carbon buyer wants to see the data, one rule holds: **institutions may receive only consented, context-preserving summaries — never raw interior reflections or ungoverned local records.** The burden is on the institution to adapt to the field's consent structure, not on the field to become legible on institutional terms. This is the practical edge of "data is offered, not extracted": legibility flows outward only as far as consent has been given, and no further.

The grammar travels; the words are translated locally

The 13×13 is a grammar, not a vocabulary. Its domain titles and prompts — and words like *Gaia*, *sacred*, *body*, *governance*, *trust*, *spirit*, AI — will not travel neutrally across Uganda, Morocco, Pakistan, Mexico, Peru, or anywhere else. Local translation is part of the method, not a courtesy added afterward. No domain title or prompt should override the language by which a community already knows its place. The grammar may travel; the words must be locally made.

When not to use this method

A method is more trustworthy for naming where it does not belong:

- Do not use it to monitor other people without consent.
- Do not use it where recording may expose sacred, protected, or vulnerable knowledge — silence is the safeguard there.
- Do not use it to satisfy a funder's reporting demand faster than the field can honestly speak.
- Do not use it to replace emergency science, professional diagnosis, legal evidence, or specialist ecological assessment where those are what is actually needed.

A claim-status table

Because the paper makes several kinds of claim, their status is set out plainly:

CLAIM	STATUS
Embodied observation is full but fallible knowledge	Philosophical / epistemological claim
The two 13×13s preserve one grammar	Structural design argument (inspectable, Part 2)
The inquiry can be practised from one bounded place or question	Operationally demonstrated
The PG Ledger can accumulate evidence over years	Architectural claim, early practice
This works at scale across cultures	Not yet empirically validated
This confers political leverage	Not claimed

6. How to begin

You bring a place, a body, and a capacity to notice. That is enough to begin inquiry. It is not, by itself, a claim of local authority, a field application, or entry into Spiralweb relationship.

The place may be a garden, courtyard, shared plot, school field, balcony, food forest, wetland edge, grazing route, street, watershed question, or another bounded relation that can be revisited. It may be smaller than one square metre or extend across many hectares. The entry discipline is not a standard size; it is an honest boundary around what you can actually observe and carry.

- Choose a bounded place or question and state why that boundary makes sense.
- Decide which prompts, if any, are useful. The eight land-based prompts in Appendix D are examples; adapt, replace, or omit them.
- Decide what must remain private, what may be recorded locally, and what could later be shared.
- Return when the place can answer: after rain, drought, planting, pruning, conflict, rest, illness, repair, or another meaningful change.
- Keep observation, inference, decision, and later outcome distinct.

- Where useful, read Land and Ecology, Steward Viability, and Coordination and Governance separately. Choose a cadence the people and place can hold.
- Use the 13×13 as a field of questions only as relevant. No cell requires completion.

No grant, platform, institutional affiliation, fixed form, or contact with the association is required. Downloading a tool or keeping a record does not create a Spiralweb application, AnchorPoint, field status, review entitlement, or funding relationship. Those arise, if at all, through slower relational and place-bound processes.

Move at Earth Time. Begin slowly enough that the field can answer. Do not force observation to satisfy a calendar or a reporting demand.

One possible learning shape — not a minimum or activation sequence:

- **Around 30 days:** revisit one bounded place and note one change or uncertainty.
- **Around 90 days:** observe through more than one condition and name one correction actually made.
- **Across a year:** compare seasonal patterns and decide, with consent, what should remain local and what may responsibly travel.

Begin with one bounded relation and one honest observation. Let the form grow only when the place, people, and purpose require it. Én dag ad gangen.

The purpose of the grammar is not to make every steward right. It is to make correction possible without removing the steward's standing.

Appendix A — Epistemological note: stratified reality and embodied knowing

This note grounds the philosophy of the method for the reader who wants it.

It draws on critical realism, principally the work of Roy Bhaskar, and offers it as scaffolding rather than scripture: one powerful and contested lens, not a neutral foundation descended from above.

Critical realism is useful here because it distinguishes the reality of the world from our knowledge of it.

Its distinction among the **real** — generative mechanisms and structures; the **actual** — events and phenomena whether observed or not; and the **empirical** — what enters experience and observation — is a reminder that reality cannot be reduced to what is presently measured, modelled or perceived.

Its warning against the **epistemic fallacy** is equally relevant.

Questions about what exists should not be collapsed into questions about how we presently know it.

But critical realism does **not** establish the categories of the 13×13 as the strata of reality.

The grid is ours.

Reality is not.

A category such as body, institution, water, economy, relationship or inner experience may direct attention toward real mechanisms, conditions or relations.

That does not make the category exhaustive, final or ontologically privileged.

The stronger critical-realist proposition is more modest:

> **Reality is stratified, emergent and independent of our descriptions; every inquiry grammar is a situated and revisable attempt to gain access to some of its relations.**

This is why qualitative and embodied observation can count as genuine evidence without becoming sufficient evidence for every question.

A lived observation can reach aspects of reality that a remote model cannot.

A hydrological test can reach dynamics that unaided embodied observation cannot.

A biodiversity survey can establish patterns that one steward's experience cannot establish.

Geology can reveal processes extending beyond a human lifetime.

Historical memory may reveal changes that a contemporary dataset misses.

Local practice may reveal consequences invisible to an externally calibrated model.

None of these facts establishes a universal hierarchy of knowing.

The question is instead:

> **What form of access does this inquiry require, what can this evidence legitimately support, and what remains outside its reach?**

“Gaia is not measured; she is met”

“Gaia is not measured; she is met” can be read in this spirit through critical realism's **intransitive/transitive distinction**.

The living Earth exists and acts independently of any one account of it.

Every observation, dataset, model, story, classification or grid belongs to the transitive dimension of knowledge: historically situated, partial, fallible and revisable.

To say that Gaia is met rather than reduced to measurement is therefore not to oppose meeting and measurement.

Measurement is one mode of meeting.

Embodied observation is another.

Scientific modelling, practical knowledge, historical memory, artistic attention, local and Indigenous knowledge — under their own authority and conditions — may provide others.

They are not interchangeable.

And none exhausts the intransitive reality toward which they reach.

The living field does not become more real when measured.

Nor does an experience become scientifically established merely because it is sincerely lived.

The purpose of the method is to keep those distinctions visible while allowing different forms of knowing to enter relation.

Judgment without a view from nowhere

This does not dissolve into “all observations are equally valid.”

Critical realism combines ontological realism with epistemic fallibility and **judgmental rationality**:

there is a world that answers back;

our accounts remain partial and revisable;

and some accounts can nevertheless be judged better than others for particular questions.

In this method, judgment is not delegated to one central verifier.

A reading may be tested through:

- repeated observation;
- material consequence;
- local review;
- comparison across time;
- specialist knowledge where relevant;
- measurement and instrumentation;
- documentary evidence;
- disagreement;
- the Correction Loop;
- and the structural separation of non-compensatory streams.

None guarantees truth.

Together they provide routes through which interpretations can become more robust or be corrected.

The purpose of correction is therefore not to replace situated knowing with a higher authority.

It is to let reality, evidence and other people answer back.

> **Correction improves knowing without requiring a view from nowhere.**

Two honest limits close the note.

First, critical realism is itself a contested philosophical framework. It is used here because its realism, emergence, fallibilism and rejection of epistemic reduction provide useful scaffolding — not because it proves this method.

Second, the practical method does not require a reader to adopt critical realism.

A person with a piece of ground, a legitimate relation to it and an honest capacity to notice is already engaged in stewardship whether or not they share this philosophy.

The grammar remains open at the level of practice.

So, deliberately, does the question of what reality finally is.

Appendix B — A brief state of the art: where this sits in citizen science

This note situates the method within the wider field, so that its claim to be different is specific rather than rhetorical. It is a short orientation, not a literature review, and it is offered in the same fallibilist spirit as the rest: a sketch open to correction.

The field, in brief. Contemporary citizen science is conventionally sorted by *degree of participation*. The widely used typology of Bonney et al. (2009), elaborated by Shirk et al. (2012), distinguishes *contributory* projects (designed by scientists; the public mainly collects or classifies data — eBird, iNaturalist, Zooniverse, distributed water- and air-quality readings), *collaborative* projects (the public also helps refine methods or interpret results), and *co-created* projects (community members involved across the whole arc, from defining the question to acting on findings — where much community-based and environmental-justice monitoring lives). A parallel tradition, *participatory action research* (Reason & Bradbury, 2008), has long insisted that those affected by a problem should help author the inquiry into it. Adjacent and increasingly important are the *Indigenous data-governance* movements — notably the CARE Principles (Carroll et al., 2020): Collective benefit, Authority to control, Responsibility, Ethics — which hold that communities must govern data about themselves and their territories; and the commons-governance scholarship of Elinor Ostrom (1990), whose design principles for self-governed common-pool resources the SRIP architecture of this series explicitly draws on.

What is genuinely strong. These movements have produced real public goods: vast biodiversity datasets that no institution could have gathered alone, early detection of ecological change, and — in the best community-monitoring cases — evidence that has shifted regulatory and legal outcomes. Participation at scale is not a small achievement, and this paper does not dismiss it.

The persistent asymmetry. Across much of the field, however, a structural tendency recurs. The typology itself is built on degree of participation in a research process whose categories, analytic pipeline, standards of validity, and data ownership still largely sit with the institution rather than the participant (Bonney et al., 2009; Shirk et al., 2012). Critics have named the recurring risks: participants positioned as distributed sensors rather than co-knowers; the broader political economy in which the capture and processing of social data becomes a form of "data colonialism," extracting from people without returning authority (Couldry & Mejias, 2019); and a validity hierarchy in which qualitative, embodied, and place-based knowledge is admitted only after translation into quantitative form. Even well-designed participatory projects often stop short of relocating *epistemic authority* — the power to define what counts as knowing. (This is offered as an observed tendency and a reading of the cited literature, not as a settled empirical finding about the whole field.)

Where this method sits, and what is distinctive. Three features distinguish the present proposal.

First, it places a **shared inquiry grammar and situated observation in direct relation** rather than treating observation as raw material for a separately sovereign analytic layer.

The 13×13 forms are homologous rather than identical, and no published grammar claims exhaustive authority over the field.

The contribution is not that one universal grid has finally been found.

It is that structured questions can sometimes travel between positions of knowing without requiring one position to own the meaning of what the others observe.

Second, the method treats **qualitative and embodied observation as knowledge in its own right** rather than as material that becomes knowledge only after quantification or expert authorisation.

This standing does not make embodied observation infallible.

It is explicitly compatible with repetition, specialist evidence, measurement, disagreement and correction.

The distinction is between **having epistemic standing** and **having exhaustive knowledge**.

Third, consent, withdrawal, protected knowledge and the witnessed/collected distinction are built into the architecture, while AI is placed under explicit corrigibility and mandate boundaries.

The purpose is to prevent distributed observation from quietly becoming a new extraction pipeline and to prevent AI-assisted synthesis from becoming the new privileged analytic authority.

The method therefore neither abolishes expertise nor installs situated experience as a replacement expertise.

It attempts something narrower:

> **to create conditions in which different forms of knowing can meet without first surrendering their differences or transferring authority automatically upward.**

This remains an experimental institutional proposition rather than an established solution to the epistemological, political or methodological problems of citizen science.

References. These anchor the orientation above; the list is selective, not exhaustive, and a fuller public version may extend it.

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- *On community-based monitoring and environmental-justice citizen science* — e.g. the health-and-environmental-justice strand surveyed in Haklay et al. (Eds.) (2020), *The Science of Citizen Science* (Springer, open access), which situates volunteer monitoring within democratic engagement and justice rather than data supply alone. https://link.springer.com/book/10.1007/978-3-030-58278-4

A *note on lineage*. The method's intellectual provenance, traced more fully in the lineage letter *Eve & Adam, and the Penguins*, is worth naming because it locates the work in a real tradition rather than presenting it as sui generis. Four strands meet here: a Danish bottom-up, cooperative methodology with roots in the andelsbevægelse and in participatory practice; the critical-realist epistemology of Appendix A; the polycentric commons-governance scholarship of Elinor and Vincent Ostrom (Ostrom, 1990), on which the SRIP architecture explicitly draws; and meta-governance theory's attention to how plural orders coordinate without a single sovereign centre. *Pixellized inquiry* (Part 2) is the methodological synthesis of these: structured, textured observation at the resolution where life actually occurs, patterned across contexts rather than abstracted from them. This is named as provenance, not validation — the lineage letter itself traces ancestry and does not stand as empirical proof of the PG Ledger, the Penguin Dashboard, or the 13×13.

Appendix C — The Gaia GoldBloom 13×13 observation grid

The 169 categories below are a map of what is possible to notice, not a checklist of what must be reported. A participant moves through them over time. No domain outranks another.

How to meet a prompt — and where to draw the line. The grid is deliberately *not* sorted into “outer, collectable” and “inner, private” columns, because that line does not run between categories — it runs through each act of observation, and where it falls is yours to decide. Any prompt here can be met in more than one way: as **material observation** (recorded, and with consent shared), as **self-witness** (noticed in your own body or life, kept or offered as you choose), or as **sacred holding** (felt and honoured, never written down at all). “Soil cover” can be a measurement or a moment of awe; “loneliness” can be a private weather you keep, or something a circle chooses to name together; “sacred water sites” may be held in silence even though they look material on the page. Deciding how you are meeting a given prompt, each time, is itself part of the practice — a small, repeatable, sovereign act, and the freedom to keep something unrecorded is as real as the freedom to record it.

One line is not yours to draw, and it is the only hard rule: **what is interior to other people is theirs to offer, never yours to collect.** Another’s grief, belief, body, or protected knowledge enters a record only when they offer it; protected, sacred, Indigenous, or community-held knowledge is governed by the people to whom it belongs (see Part 5). In some cases, even the *presence* of such knowledge or such a site should not be recorded — placement, existence, ritual timing, or community relation may themselves be protected. Silence is a valid form of protection.

These prompts are not diagnostic instruments, investigation mandates, or licences to assess other people. Prompts like *corruption signals*, *trauma awareness*, *mental health support*, *loneliness*, or *community trust* are invitations to notice conditions — beginning with one’s own position — and to record only what is material, offered, or properly governed. Within that floor, the placing of every other line is yours — and the quiet thing worth noticing is that drawing it for yourself, rather than having it drawn for you, is already a small healing of the split between data and feeling that this method exists to refuse.

1. Earth Systems Sensing 1. Soil vitality observation · 2. Water clarity & flow · 3. Air quality felt-sense · 4. Biodiversity presence logs · 5. Seasonal rhythm tracking · 6. Weather anomaly noticing · 7. Mycelial activity awareness · 8. Plant phenology notes · 9. Pollinator encounters · 10. Erosion & regeneration signals · 11. Temperature micro-zones · 12. Night sky visibility · 13. Earth soundscapes

2. Body as Sensor 1. Heart coherence · 2. Breath rhythm · 3. Muscle tone changes · 4. Fatigue signals · 5. Joy resonance · 6. Stress thresholds · 7. Digestive feedback · 8. Sleep quality · 9. Sensory overload detection · 10. Grounding capacity · 11. Movement ease · 12. Pain signals · 13. Embodied intuition

3. Water Guardianship 1. Drinking water taste · 2. Local stream health · 3. Rainwater patterns · 4. Flood memory mapping · 5. Drought noticing · 6. Watershed identity · 7. Wastewater awareness · 8. Ocean mood sensing · 9. Ice & snow presence · 10. Fog and humidity · 11. Sacred water sites · 12. Water rituals · 13. Hydrological grief & joy

4. Food & Soil Commons 1. Seed saving stories · 2. Soil smell index · 3. Crop diversity counts · 4. Urban garden health · 5. Foraging notes · 6. Food access equity · 7. Composting rhythms · 8. Microbial awareness · 9. Local food webs · 10. Seasonal hunger patterns · 11. Taste memory · 12. Food sovereignty signals · 13. Gratitude practices

5. Human Ecology 1. Community trust levels · 2. Conflict presence · 3. Care networks · 4. Loneliness signals · 5. Intergenerational contact · 6. Ritual frequency · 7. Shared meals · 8. Local leadership health · 9. Migration stories · 10. Belonging index · 11. Mutual aid flows · 12. Cultural grief · 13. Collective joy

6. Built Environment 1. Shelter quality · 2. Thermal comfort · 3. Light access · 4. Noise pollution · 5. Walkability · 6. Material toxicity · 7. Repair culture · 8. Public space vitality · 9. Infrastructure decay · 10. Sacred architecture · 11. Temporary structures · 12. Home adaptability · 13. Resilience capacity

7. Energy & Flow 1. Energy access · 2. Grid reliability · 3. Renewable presence · 4. Energy fatigue · 5. Heat stress · 6. Cold exposure · 7. Movement of people · 8. Movement of goods · 9. Data flow awareness · 10. Rest cycles · 11. Overproduction signals · 12. Flow blockages · 13. Regenerative surplus

8. Climate Emotion 1. Eco-anxiety · 2. Climate grief · 3. Hope signals · 4. Denial presence · 5. Anger waves · 6. Apathy zones · 7. Love for place · 8. Fear thresholds · 9. Resilience stories · 10. Imagination capacity · 11. Future orientation · 12. Mourning rituals · 13. Courage moments

9. Learning & Knowledge 1. Local wisdom · 2. Indigenous knowledge presence · 3. Peer learning · 4. Formal education gaps · 5. Skill sharing · 6. Story transmission · 7. Language loss · 8. Curiosity signals · 9. Learning joy · 10. Unlearning processes · 11. Mistake culture · 12. Knowledge commons · 13. Epistemic humility

10. Technology & AI 1. Tool usefulness · 2. Digital overwhelm · 3. Access equity · 4. Automation stress · 5. Human-in-the-loop signals · 6. Transparency · 7. Energy cost of tech ·

8. Repairability · 9. Surveillance presence · 10. Consent awareness · 11. Open-source vitality · 12. AI trust signals · 13. Tech as ally or burden

11. Governance & Power 1. Decision transparency · 2. Participation levels · 3. Corruption signals · 4. Local autonomy · 5. Rule legitimacy · 6. Consent felt-sense · 7. Enforcement harm · 8. Care-based governance · 9. Emergency powers · 10. Commons protection · 11. Voice inclusion · 12. Silencing patterns · 13. Power regeneration

12. Health & Care 1. Access to care · 2. Preventive culture · 3. Chronic illness signals · 4. Mental health support · 5. Community healers · 6. Medical debt · 7. Caregiver load · 8. Rest permission · 9. Birth & death rituals · 10. Disability inclusion · 11. Healing environments · 12. Trauma awareness · 13. Whole-person care

13. Meaning, Spirit & Future 1. Sense of purpose · 2. Ritual presence · 3. Sacred time · 4. Hope for children · 5. Cosmology diversity · 6. Prayer & silence · 7. Grief integration · 8. Mystery tolerance · 9. Beauty noticing · 10. Art as signal · 11. Time horizon depth · 12. Interbeing awareness · 13. Commitment to life

Closing note. This 13×13 citizen-science framework is intentionally qualitative, relational, and slow. Data here is not extracted but offered. Meaning arises through patterning across bodies, places, and times. Participation is voluntary, sovereign, and grounded in care — a meeting with the living world, not a measurement of it.

Appendix D — Example place-bound observation sheet (one page)

This is one illustrative land-based sheet, not a universal record. Copy it into a notebook only where the prompts fit; adapt, replace, or remove columns as the place, purpose, and local knowledge require. One row may represent a date, event, season, or another locally meaningful return.

> **This sheet belongs first to the steward and local circle.** Do not send, upload, photograph, or share it without explicit consent. Use initials, codes, or approximate location where privacy matters.

Bounded place / question: ________________ **Steward / local holder:** ________________ **Location / scale / privacy code:** ________________

(Use an approximate location or a local code if precise location should not travel — see the sacred/protected-knowledge rule in Part 5.)

DATE	1 SOIL COVER	2 VEGETATION LAYERING	3 WATER BEHAVIOUR	4 SUCCESSION	5 BIODIVERSITY	6 SOIL HEALTH	7 BIOMASS CYCLE

How to use it. The eight columns are example prompts, not required indicators. Replace them where another local grammar is more truthful. Use a few words, a drawing, a code, or another suitable form. Human rhythm is self-witnessed, never collected about others. A blank or an unrecorded domain is legitimate. Distinguish *I noticed*, *I infer*, *we decided*, and *what happened next*. The record is not for proving activation or success; it is for improving judgement.

Appendix E — Example three-stream review (one page)

Read at a locally useful interval. Monthly review is one option. Mark each stream green / yellow / red **separately** — never average them into one score. The separation is the point.

Review period / event: ________________ **Place / application:** ________________ **People holding the review:** ________________

STREAM	GREEN	YELLOW	RED	THIS REVIEW
A — Land & Ecology	cover, water-holding, biodiversity, or soil improving	mixed signals, stagnation, uncertainty	erosion, die-off, compaction, contamination, harm	☐ G ☐ Y ☐ R
B — Steward Viability	rhythm sustainable	strain visible; reduce ambition	depletion, conflict, dread, illness, hidden labour; pause	☐ G ☐ Y ☐ R
C — Coordination & Governance	roles & decisions clear	confusion, informal overload	conflict, coercion, unclear consent, untraceable decisions	☐ G ☐ Y ☐ R

Guardrail. If Stream B is red, pause Stream A: ecological ambition must not be financed by human depletion. A green in one stream never cancels a red in another.

> **Red is not failure. Red is information.** A red stream protects the field, the steward, or the circle from false success. Hiding red to preserve a success story is the one move this sheet exists to prevent.

One question to close. Is this giving to life? Answer honestly, even when the answer is not yet green.

Related architecture

This paper draws on a wider body of work. A reader needs only the terms introduced in the body — situated knowing, a shared inquiry grammar, provisional observation prompts, non-compensatory streams, the Last Impulse, and the Moral Biology grounding. The full living index is at **https://papers.spiralweb.earth**.

Required for this paper

- **Moral Biology** — Green Paper 01, the conceptual ground: ethics as capacity, originating in bodies.

- **13×13 inquiry grammar / The Correction Loop** — Report 02 provides one reflective 13×13 lineage and the dedicated accountability mechanism for consequential AI-assisted work. Earlier "Planetary Operating System" language belongs to the development history rather than the current architectural description.
- **SRIP — The Steward's Journey** — Report 05: bounded entry with a real place, light observation, the three-stream distinction, burden testing, relational pathways, pause, withdrawal, and dignified incompletion. Ten square metres, eight prompts, and a Circle of 13 are possible forms, not universal requirements.
- **The Protocol Habitat** — the current architecture distinguishing Foundations, Protocols, Applications, and Tools; publication from activation; relationship from local authority; and generic material from place-bound application.
- **PG Ledger** — a place-bound memory, flow, evidence, and correction architecture. It may hold observations, burdens, decisions, consent states, financial flows, histories, and review without collapsing them into one score. No fixed category list, cadence, or tool is active across all fields.

Operational companions

- **Operational Formats** — printable candidate tools for dashboard reading, observation, bounded action, decision memory, and financial traceability. Tools may be adapted or left unused; they do not certify field truth, create local authority, or activate a field.
- **Penguin Dashboard** — Report 04: a human governance reading that keeps Land and Ecology, Steward Viability, and Governance and Collaboration non-compensatory.
- **Circle of 13** — one possible human-scale governance form where locally chosen; not a required number or entry condition.
- **AnchorPoint** — the relational, place-based threshold where local continuity may meet the wider architecture; it holds relationship, not automatic application or authority. https://spiralweb.earth/cafes/sofia/anchorpoints/
- **Pixellized inquiry** — a possible methodological bridge from bounded situated observation to wider patterning, provided context and correction remain attached.

Wider horizon *(context, not prerequisite)*

- **Rule of Life** — the constitutional horizon: no economy, institution, technology, right, or legal form may override the conditions of life, while all action remains bound to rule of law, evidence, procedure, review, and proportionality.
- **Earth Time** — the temporal discipline that living systems set the pace.

- **Gaia GoldBloom / Gold Before Bloom** — Green Paper 11, the ecosystem and its stabilise-before-expanding logic.
- **Penguin Economics / Regenerative Reciprocity** — the wider economic and flow architecture: rotation, anti-hoarding, viable circulation, support without capture, and abundance circulating only after life is stable enough to give.
- **Sophia Lumen** — the living relational human–AI practice of co-inquiry; not a persona, autonomous author, or field authority.
- **Sophia Lumen Protocol** — the current candidate protocol for human–AI co-inquiry: context cultivation, provenance, prismatic inquiry, structural asymmetry, temporary actionability, and recursive correction.
- **Methods / Editorial Practice** — how the Green Papers are written, revised, and held.

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